Fenchel Dual

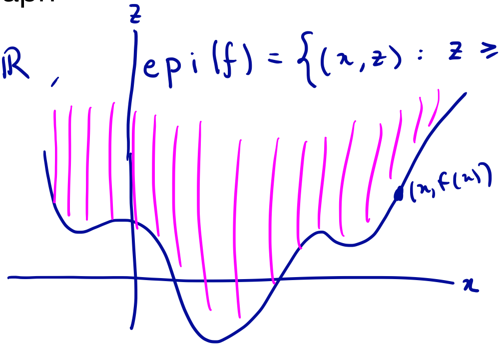
$$f : \mathbb{R}^n \to \mathbb{R}$$

$$f^*(y) \doteq \sup_x \{ y^T x - f(x) \}$$

We saw: if f is quadratic then ∇f^* is the inverse of ∇f.

The Epigraph

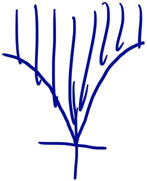
$f : \mathbb{R}^n \to \mathbb{R}$, $\quad \text{epi}(f) = \{(x, z) : z \geq f(x)\}$



Exercise: epi(f) is convex iff f is convex.

Note: level sets of f are horiz. slices of epi(f).

epi f cvx $\Rightarrow$ level sets are convex. But:



The Conjugate is Closed and Convex

Recall: a set $C \subseteq \mathbb{R}^n$ is closed if it contains all of its limit points. That is: $\{x_n\} \subseteq C$, $x_n \to x \Rightarrow x \in C$

Check: (arbitrary) intersection of closed sets is closed

finite union of closed sets is closed, (but an arbitrary union may not be).

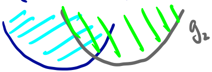
Any affine map of a closed set is closed.

(In fact: inverse of any continuous map maps closed sets to closed sets)

Def'n: A fn $f : \mathbb{R}^n \to \mathbb{R}$ is called **closed** if $\operatorname{epi}(f) \subseteq \mathbb{R}^{n+1}$ is closed.

Fact: For $f : \mathbb{R}^n \to \mathbb{R}$ any function, $f^*(y) = \sup \{ y^T x - f(x) \}$ is closed and convex.

Pf: Convexity: $f^*(y)$ is sup. of affine fns hence convex.



Closed: Note: $g(x) \triangleq \max\{g_1(x), g_2(x)\}$

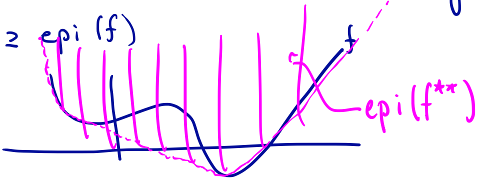
$\operatorname{epi} g = \operatorname{epi} g_1 \cap \operatorname{epi} g_2$. Hence: f^* is **closed**.

Conjugate of the Conjugate

$f : \mathbb{R}^n \to \mathbb{R}$, $\quad f^*(y) = \sup\{y^T x - f(x)\}$, $f^{**}(x) = \sup_y \{x^T y - f^*(y)\}$

<u>Check</u> : $\operatorname{epi}(f^{**}) \supseteq \operatorname{epi}(f)$

Illustration:

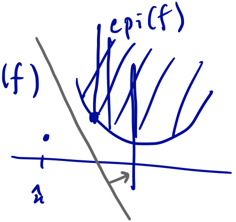


If f is closed and convex, then $f^{**} = f$.

We will show: $\operatorname{epi}(f^{**}) = \operatorname{epi}(f)$.

We will do this by contradiction.

Conjugate of the Conjugate - proof -



Suppose $\exists\, \hat{x}$ with $(\hat{x}, f^{**}(\hat{x})) \notin \text{epi}(f)$

$$H_{\tilde{\alpha},\beta} = \{ z : \tilde{\alpha}^T z = \beta \},$$

$\tilde{\alpha} \in \mathbb{R}^{n+1}$, $\quad \tilde{\alpha} = (\alpha, \alpha_0) \in \mathbb{R}^n \times \mathbb{R}$

$\beta \in \mathbb{R}$

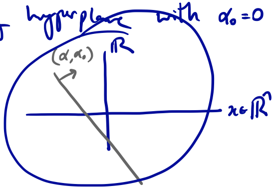
$H_{\tilde{\alpha},\beta}$ separates $(\hat{x}, f^{**}(\hat{x}))$ from epi(f).

$$\alpha^T x + \alpha_0 s - (\alpha^T \hat{x} + \alpha_0 f^{**}(\hat{x})) \leq \beta < 0 \quad \forall (x, s) \in \text{epi}(f)$$

Note: $\alpha_0 \leq 0$ (o.w. take $s \to +\infty$ for a contradiction)

Conjugate of the Conjugate - proof - continued -

Suppose $\alpha_0 = 0$. $\longrightarrow$ What does a separating hyperplane with $\alpha_0 = 0$ look like?



Let $\hat{y} \in \operatorname{dom} f^*$

Consider: $H_{(\underbrace{\tilde{\alpha} + \varepsilon(\hat{y},-1)}_{\text{new normal vector}},\ \beta)}$ $\qquad (\tilde{\alpha} = (\alpha, \overset{\alpha_0}{0}))$

$$\alpha^T x + \cancel{\alpha_0 s} - (\alpha^T \hat{x} + \cancel{\alpha_0 f^{**}}(\hat{x})) \le \beta < 0 \quad \forall (x,s) \in \operatorname{epi}(f)$$

$$(\alpha + \varepsilon\hat{y})^T x - \varepsilon s - ((\alpha+\varepsilon\hat{y})^T \hat{x} - \varepsilon f^{**}(\hat{x})) \le \underbrace{\beta + \varepsilon(\hat{y}^T x - s) - \hat{y}^T\hat{x} + f^{**}(\hat{x}))}\ \overset{?}{<} 0 \quad \forall (x,s)$$

maximize over $(x,s) \in \operatorname{epi}(f)$ $\in \operatorname{epi}(f)$

$$\le \beta + \varepsilon\left(f^*(\hat{y}) - \hat{y}^T\hat{x} + f^{**}(\hat{x})\right)$$

< 0 for ε small enough.

Upshot: If $\exists$ pt. $(\hat{x}, f^{**}(\hat{x})) \notin \operatorname{epi}(f) \Rightarrow \exists$ sep. hyperplane with $\alpha_0 < 0$.

$$\alpha^T x - \alpha^T \hat{x} + \alpha_0 s - \alpha_0 f^{**}(\hat{x}) \le \beta < 0 \quad \forall (x,s) \in \operatorname{epi} f$$

$$\Rightarrow (\alpha/-\alpha_0)^T x + f^{**}(\hat{x}) - (\alpha/-\alpha_0)^T \hat{x} - s \le \beta/-\alpha_0 \longrightarrow \forall (x,s) \in \operatorname{epi} f$$

Take sup over $(x,s) \in \operatorname{epi}(f)$

Then: $f^*(y) + f^{**}(\hat{x}) - y^T\hat{x} \le \beta/-\alpha_0 < 0$: ① What is y? ② Why is this a contradiction.